

📌 Quick Recall (Ch. 7.2)

Set systems & antichains (Def). For a finite set X with $|X| = n$, a *set system* $\mathcal{M} \subseteq 2^X$ is called *independent* (or an *antichain*) if it contains no two distinct sets A, B with $A \subset B$.

Chains. A *chain* in 2^X is a family $\{A_1, \dots, A_k\}$ with $A_1 \subset A_2 \subset \dots \subset A_k$. A *maximal chain* adds elements one at a time:

$$\emptyset \subset \{x_1\} \subset \{x_1, x_2\} \subset \dots \subset X.$$

Maximal chains are in bijection with linear orderings of X – exactly $n!$ of them.

Key observation. Any chain meets any antichain in at most one set.

Sperner's Theorem (Thm 7.2.1). Every antichain $\mathcal{M}$ on an n -element set satisfies

$$|\mathcal{M}| \leq \binom{n}{\lfloor n/2 \rfloor}.$$

Equality is attained by $\mathcal{M} = \binom{X}{\lfloor n/2 \rfloor}$.

Proof sketch (chain counting). Count pairs (R, M) with $M \in \mathcal{M}$ and R a maximal chain through M :

- Each chain meets $\mathcal{M}$ at most once $\Rightarrow \#(R, M) \leq n!$.
- Each $M \in \mathcal{M}$ lies on $|M|!(n - |M|)!$ maximal chains.

Combining gives the LYM inequality

$$\sum_{M \in \mathcal{M}} \binom{n}{|M|}^{-1} \leq 1,$$

hence $|\mathcal{M}| \leq \binom{n}{\lfloor n/2 \rfloor}$.

📌 Practice: Textbook Exercise 7.2.6* — the Littlewood–Offord problem

Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ with $|a_i| \geq 1$ for every i . For each sign vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{-1, +1\}^n$, define

$$S(\varepsilon) := \sum_{i=1}^n \varepsilon_i a_i, \quad p(a_1, \dots, a_n) := \#\{\varepsilon \in \{\pm 1\}^n : -1 < S(\varepsilon) < 1\}.$$

(a)* Prove that for any choice of $a_1, \dots, a_n$ with $|a_i| \geq 1$,

$$p(a_1, \dots, a_n) \leq \binom{n}{\lfloor n/2 \rfloor}.$$

(b) Find $a_1, \dots, a_n$ with $p(a_1, \dots, a_n) = \binom{n}{\lfloor n/2 \rfloor}$.

💡 Hints

Hint for (a) (textbook). Apply Sperner's theorem on the set system $\{\{i : \varepsilon_i = 1\} : \sum_i \varepsilon_i a_i \in (-1, 1)\}$.

Suggested steps. (1) WLOG assume $a_i > 0$ for all i (use a sign-flip bijection on $\{\pm 1\}^n$). (2) For each valid ε , set $S_\varepsilon := \{i : \varepsilon_i = +1\}$. (3) Show $\mathcal{M} := \{S_\varepsilon : \varepsilon \text{ valid}\}$ is an antichain by computing $S(\delta) - S(\varepsilon)$ when $S_\varepsilon \subsetneq S_\delta$.

Hint for (b). Try the simplest possible choice: $a_1 = a_2 = \dots = a_n = 1$. For which parity of n does this attain the bound, and why?

Answer: Part (a).

Step 1: WLOG $a_i \geq 1$ (all positive). For each j with $a_j < 0$, define $\phi_j : \{\pm 1\}^n \rightarrow \{\pm 1\}^n$ by flipping only the j -th coordinate, $\phi_j(\varepsilon) := (\varepsilon_1, \dots, -\varepsilon_j, \dots, \varepsilon_n)$; this is a bijection (in fact an involution). Setting $a^{(j)} := (a_1, \dots, |a_j|, \dots, a_n)$, a direct computation gives

$$S(\phi_j(\varepsilon); a) = \sum_{i \neq j} \varepsilon_i a_i + (-\varepsilon_j) a_j = \sum_{i \neq j} \varepsilon_i a_i + \varepsilon_j |a_j| = S(\varepsilon; a^{(j)}).$$

So ε is valid under $a^{(j)}$ iff $\phi_j(\varepsilon)$ is valid under a , and bijectivity of ϕ_j yields $p(a) = p(a^{(j)})$. Iterating over all j with $a_j < 0$, we may assume $a_i \geq 1$ for all i .

Step 2: Encode each valid ε by a subset. Call ε *valid* if $-1 < S(\varepsilon) < 1$. For each valid ε , set

$$S_\varepsilon := \{i : \varepsilon_i = +1\} \subseteq [n].$$

The map $\varepsilon \mapsto S_\varepsilon$ is a bijection $\{\pm 1\}^n \leftrightarrow 2^{[n]}$, so $p = |\mathcal{M}|$ where $\mathcal{M} := \{S_\varepsilon : \varepsilon \text{ valid}\}$.

Step 3: $\mathcal{M}$ is an antichain. Suppose for contradiction $S_\varepsilon \subsetneq S_\delta$ for two valid sign vectors ε, δ . Let $T := S_\delta \setminus S_\varepsilon \neq \emptyset$.

- For $i \in T$: $\delta_i = +1$ and $\varepsilon_i = -1$, so $\delta_i - \varepsilon_i = 2$.
- For $i \notin T$: either $i \in S_\varepsilon \subseteq S_\delta$ (both $+1$) or $i \notin S_\delta$ (both -1); in either case $\delta_i = \varepsilon_i$.

Therefore

$$S(\delta) - S(\varepsilon) = \sum_{i=1}^n (\delta_i - \varepsilon_i) a_i = \sum_{i \in T} 2a_i \geq 2|T| \geq 2,$$

using $a_i \geq 1$ and $|T| \geq 1$. But both $S(\varepsilon), S(\delta) \in (-1, 1)$ force $|S(\delta) - S(\varepsilon)| < 2$. $\nmid$

Step 4: Apply Sperner's theorem. Since $\mathcal{M}$ is an antichain in $2^{[n]}$, Sperner gives $|\mathcal{M}| \leq \binom{n}{\lfloor n/2 \rfloor}$, hence $p(a_1, \dots, a_n) = |\mathcal{M}| \leq \binom{n}{\lfloor n/2 \rfloor}$. $\square$

Part (b). Let n be *even* and take $a_1 = a_2 = \dots = a_n = 1$. Then for every ε ,

$$S(\varepsilon) = \sum_i \varepsilon_i = 2|S_\varepsilon| - n,$$

so $-1 < S(\varepsilon) < 1 \iff |S_\varepsilon| = n/2$. The number of $\varepsilon \in \{\pm 1\}^n$ with $|S_\varepsilon| = n/2$ is exactly $\binom{n}{n/2} = \binom{n}{\lfloor n/2 \rfloor}$. $\square$

Remark (for the curious). For n odd, $\sum_i \varepsilon_i$ is always odd, so $a_i = 1$ never produces $S(\varepsilon) \in (-1, 1)$ at all – under the strict-inequality formulation, the bound is in fact *not* attained for n odd. The original Littlewood–Offord theorem (Erdős, 1945) considers the count of ε with $S(\varepsilon)$ in any open interval of length 2, and the bound $\binom{n}{\lfloor n/2 \rfloor}$ is tight for all n .